

EXCLUSIVE

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GS-1 (DYNAMIC)

MODULE - 66



**SCIENCE
&
TECHNOLOGY-14**

What do you know about India's first significant antibiotic innovation in 30 years?



- Wockhardt Laboratories has developed a brand-new antibiotic named “Zaynich” intended specifically to tackle a class of multidrug resistant bacteria known as Gram-Negative Bacteria. Zaynich merges two different antibiotics, cefepime with zidebactam.
- This development is being hailed as India's first major antibiotic innovation in about three decades.
- Examples of gram-negative bacteria are *E. coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, and *Acinetobacter* species. These are notoriously difficult to treat because their cell structures and resistant enzymes can block many standard drugs. These powerful bacteria cause major health issues like pneumonia and meningitis.
- If proven safe and effective in broader clinical use, Zaynich could become a tool both domestically and globally, helping doctors manage infections when other antibiotics fail.



- Electric vehicle batteries/energy storage batteries require higher energy density, safety, temperature resilience and environmental sustainability.
- As per the emerging research, it is demonstrated that liquefied gas electrolytes can simultaneously provide all four essential properties.
- The screened Fluoromethane (FM) and Difluoromethane (DFM) small molecules have a low melting point, fast kinetic properties and wide voltage window.
- These characteristics allowed these LGEs to exhibit excellent low temperature performance (lesser than -60°C), Lithium Metal Coulombic Efficiency (greater than 99.8%) and to work with high performance, high voltage cathodes.



As of mid-2025, 47 countries or territories have been officially certified malaria-free by the WHO.

INDIA TARGETS ELIMINATION OF MALARIA BY 2030

- In 2016, under its National Framework for Malaria Elimination in India (2016-2030), India set an ambitious target to eliminate malaria (zero indigenous cases) by 2030, with an interim milestone of interrupting indigenous transmission across the entire country, including all high-transmission States and UTs by 2027.
- By the end of 2025, the Ministry of Health and Family Welfare (MOHFW) reported that robust surveillance and sustained interventions had led to 160 districts across 23 States and UTs reporting zero indigenous malaria cases from 2022 to 2024. This was seen as a significant milestone as it meant that the country was getting closer to completely eliminating malaria.

HOW IS THE PREVALENCE OF MALARIA MEASURED?

As per WHO, a country is granted a certification of malaria elimination when “the chain of local transmission have been interrupted nationwide for at least three consecutive years, and that a fully functional surveillance and response system is in place to prevent re-establishing of indigenous transmission”.



WHERE DOES INDIA STAND?

- As per World Malaria Report 2025 by WHO, India made significant progress in reducing malaria incidence and mortality in its high-endemic states.
- Officially, India exited the WHO “High Burden to High Impact” Group, in 2024. Malaria cases reduced by around 80% from 2015 to 2023 in the country.
- India is on track to achieve the WHO Global Technical Strategy (GTS) for malaria, 2016-2030 target of at least a 75% reduction in incidence by 2025 (compared to 2015 baseline).

As per the World Malaria Report, India already achieved reductions exceeding 70% by 2024.



India has put in place two national plans to guide and accelerate malaria elimination - the National Framework for Malaria Elimination in India (2016-2030), which outlines the vision, goals, and targets for a phased malaria elimination, and the National Strategic Plan (NSP) for Malaria Elimination (2023-2027) that builds upon earlier frameworks. As per NSP

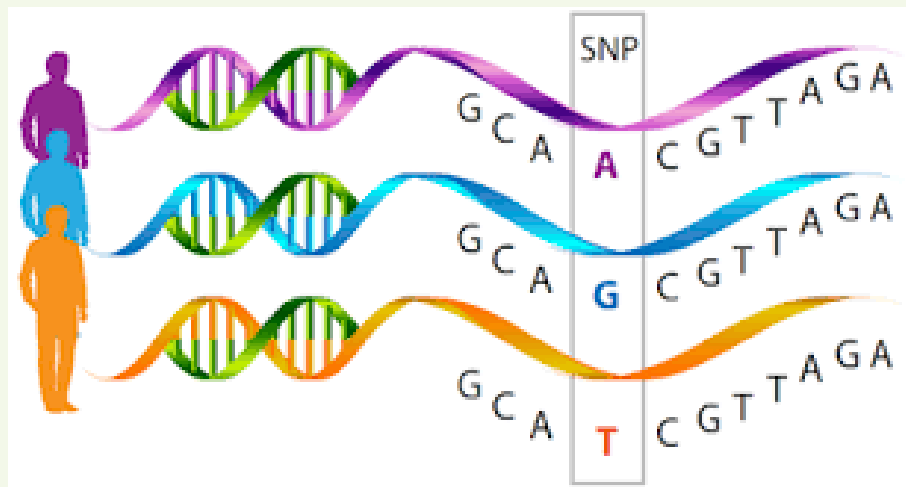
- ✓ Transforming malaria surveillance as a core intervention for malaria elimination
 - ✓ Ensuring universal access to malaria diagnosis
 - ✓ Treatment by enhancing and optimising case management by “testing, treating and tracking”
 - ✓ Ensuring universal access to malaria prevention by enhancing and optimising vector control
- are among the key strategies.

What do you know about the indigenous 4G stack developed by India?



- The indigenously developed 4G technology stack was unveiled recently by the Prime Minister.
- For the stack, the Center for Development of Telematics developed the core network, Tejas Networks provided the Radio Access Network (RAN) and Tata Consultancy Services served as the systems integrator.
- Besides, India, four countries - China, South Korea, Denmark, and Sweden - have developed a 4G mobile technology stack.
- For users, a 4G stack would mean faster and reliable mobile internet services and for the country, the indigenous technology is expected to offer the benefits of national security and digital sovereignty.
- Another advantage of the 4G stack is a possible upgrade to the 5G infrastructure in future because of the software-first design of the platform.

The homegrown 4G stack, deployed for Bharat Sanchar Nigam Limited, is being readied for export as well.



Research is ongoing to identify SNPs associated with complex diseases such as heart disease, diabetes and cancer.

- During the Single Nucleotide Polymorphisms (SNP) Test, a sample of DNA is isolated from the blood sample. Then scientists use this sample to identify genetic variations, known as Single Nucleotide Polymorphisms (SNPs).
- These SNPs tell scientists whether the body will react positively or not to a specific treatment.
- It provides essential information on how the body will respond to treatment with common anti inflammatories, antibiotics and anticoagulants.
- SNPs can also be used to track the inheritance of disease-associated genetic variants within families.

SNPs can act as biological markers helping scientists locate genes that are associated with the disease.



- A QR code consists of black squares arranged in a square grid on a white background.
- In 1994, a Japanese Company called Denso Wave invented the QR Code, which was used to label car parts.
- The idea was to replace the numerous bar code labels that had to be scanned on each box of auto parts with a single label.
- The International Organization for Standardization (ISO) added the QR Codes to their list in the year 2000, giving it International Certification.
- With the development of smartphones, QR Codes' popularity increased tremendously.

DENSO
DENSO WAVE

Using QR Codes to access websites, networks and payment details is the quickest method. Someone has to simply scan the code and do not need to enter any URL.



- About 92 inmates of Jalpaiguri Central Correctional Home (JCCH) in West Bengal were infected with the Herpes Simplex Virus during the past six months with seven of the infected inmates losing their lives due to the infection.
- While HSV is a common virus, if untreated, it can lead to encephalitis, a serious condition involving inflammation of the brain, and can turn fatal.
- The number of inmates at the correctional home with a capacity of 700 is over 1,200. As per Chief Medical Officer, overcrowding is one of the biggest risks for the HSV outbreak. Close physical contact, shared utilities and poor personal hygiene were among the other points pointed out by the CMO.
- The steps suggested to control the virus included isolation and cohorting of infected cases, improving personal and hand hygiene, better ventilation and decongesting the barracks.

As per Prison Statistics of India 2023, district prisons in West Bengal have reported an occupancy rate of 158%.

What do you know about the recent discovery of uranium content in breast milk?



- Forty randomly selected samples from various districts of Bihar were tested between October 2021 and July 2024. The results showed that all breast milk samples had uranium concentrations between 0 microgram per L and 5.25 microgram per L.
- The authors linked uranium in breast milk to uranium contamination of groundwater citing earlier groundwater studies to establish the link.
- There is currently no permissible limit or benchmark specified for uranium concentration in breast milk. However, the World Health Organization has set a provisional guideline limit of 30 microgram per liter of uranium in drinking water.
- Hence, the levels observed are expected to have minimal actual health impact on mothers and infants, and the breastfeeding should continue.

Some experts said, while reporting such studies, the authors must be cautious.



- Hayli Gubbi volcano in Ethiopia recently erupted for the first time in nearly 12,000 years, sending thick plumes of ash up to 14 km into the sky. Moving at around 100 to 120 km per hour, the plume carried volcanic ash, sulfur dioxide, and tiny particles of glass and rock.
- A jet engine works by sucking in air, squeezing it, mixing it with fuel, burning it and pushing hot gases out the back to produce thrust.
- The engine runs extremely hot at temperatures of 1,600 degrees Celsius, while running very fast with air racing through it at 600 miles per hour.
- When volcanic ash gets into this environment, the silicate components melt and re-solidify at high temperatures, creating a glassy deposit on hard parts which blocks tiny cooling holes, choking off flow within the engine. This can lead to engines losing power or shutting down.

After lying silent for at least 12,000 years, a shield volcano, Hayli Gubbi erupted on November 23. It is in Ethiopia's Erta Ale range in the geologically active Rift Valley. The African and Arabian plates under Hayli Gubbi are slowly moving apart at roughly 0.4 to 0.6 inches a year.

What do you understand by Functional Foods? What are Smart Proteins?



- Functional foods are enriched foods that promote health or prevent disease, such as vitamin-enriched rice or omega-3 fortified milk. Functional foods leverage several technologies, such as nutrigenomics (the study of how nutrition interacts with genes), biofortification, 3D foodprinting, and bioprocessing.
- Smart proteins are referred to proteins sourced using biotechnology that aim to reduce reliance on conventional production. These include plant-based proteins (restructured extracts from legumes, cereals, or oil seeds to mimic animal meat and dairy), fermentation-derived proteins (produced by microbial systems) and cultivated meat (animal cells grown in bioreactors without the slaughter).



- Biochemical markers are small molecules produced during metabolic processes in the body. These include sugars, amino acids, lipids, and other compounds that reflect the state of various organisms and systems.
- Metabolomic profiling, the large-scale study of these molecules, allows researchers to detect subtle changes in the body's chemistry that may precede disease. Using advanced techniques like liquid chromatography and mass spectrometry, scientists can analyze hundreds of metabolites from a single blood sample.
- A new study published by the researchers from IIT Bombay, Osmania Medical College, and Clarity Biosystems suggests that these tiny molecules could serve as early warning signals for diabetes and its complications, particularly kidney disease.
- Early detection tools, like metabolomic profiling, could be transformative. Healthcare providers can intervene earlier, potentially preventing complications such as kidney failure, heart disease, and nerve damage. Moreover, Metabolomic profiling could enable personalized care. Patients could be grouped based on their metabolic profiles, allowing doctors to tailor interventions from lifestyle changes to medication based on individual risk.

What do you know about the recent launch of the Research Development and Innovation (RDI) Fund?

On July 1, 2025, the Union Cabinet approved an initiative to set up Rs. 1 lakh crore RDI fund aimed at spurring a private sector-driven R&D ecosystem.



- The Executive Council of the Anusandhan National Research Foundation recently approved the establishment of a special purpose fund for the RDI scheme as also the implementation guidelines and the governance framework.
- The implementation guidelines and the special financial rules are prepared by the Department of Science and Technology. The Ministry of Science and Technology is the nodal ministry for the scheme.
- RDI will operate through a two-tiered funding structure. At the first level, a Special Purpose Fund will be established within the National Anusandhan Research Foundation, which will serve as the custodian of the Rs. 1 lakh crore corpus. The fund will not directly invest in industries and startups, but instead channel capital to second-level fund managers, which can be alternative investment funds, development finance institutions, or NBFCs, among others.
- Recommendations for support will be made by second-level fund managers through investment committees comprising experts from the financial, business, and technical domains, operating at arm's length from the government.

What do you know about the first Emerging Science, Technology and Innovation Conclave (ESTIC)?



- The Prime Minister recently inaugurated the first Emerging Science, Technology and Innovation Conclave (ESTIC).
- He stated that India's expenditure on R&D had doubled in the previous decade. The number of patents registered had grown 17 times and the number of deeptech startups risen to 6,000, as per the Prime Minister.
- He further said, India had made operational a fund of Rs. 1 lakh crore via the Anusandhan National Research Foundation, which would help scientists and technologists in the public and private sectors to invest more substantially in research and development.

The ESTIC replaces the Indian Science Congress, the oldest congregation of scientists in India. Indian Science Congress had gone into oblivion with its last session held in 2023.

What do you know about the recently announced Rashtriya Vigyan Puraskar awards?



- Recently, the Centre announced the Rashtriya Vigyan Puraskar (RVP) awards for scientific achievement with twenty-four individual awards and one team award.
- Eminent physicist Jayant Vishnu Narlikar, who passed away in May 2025, was announced the recipient of the Vigyan Ratna, an award for lifetime contribution.
- Eight scientists were selected for the Vigyan Shri awards. The Vigyan Team award went to the CSIR Aroma Mission.
- An independent committee - the Rashtriya Vigyan Puraskar Committee, steered by the Principal Scientific Adviser - decides on award recipients and makes a recommendation to the Minister of Science and Technology. It is unclear if the Minister has a veto on the committee's recommendations.
- In theory, the total number of awards under all categories is capped at 56. There is no cash component given for the RVP.

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